

	Work Instruction	Doc. Revision	00
	AXI V810 S2EX Rail Width Adjustment and Verification	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☐	S1	☐	S2
V810 S2EX	☒			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	2 Oct 2016	First Release	ONG KEAN SHENG

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Rail Width Adjustment and Verification

1. Initialize at Panel Handling
2. Set the Display Measurement to millimeters as below Figure 1
3. Go to the Set Manual Adjustment as key in 350 and click "Set" as below Figure 1.

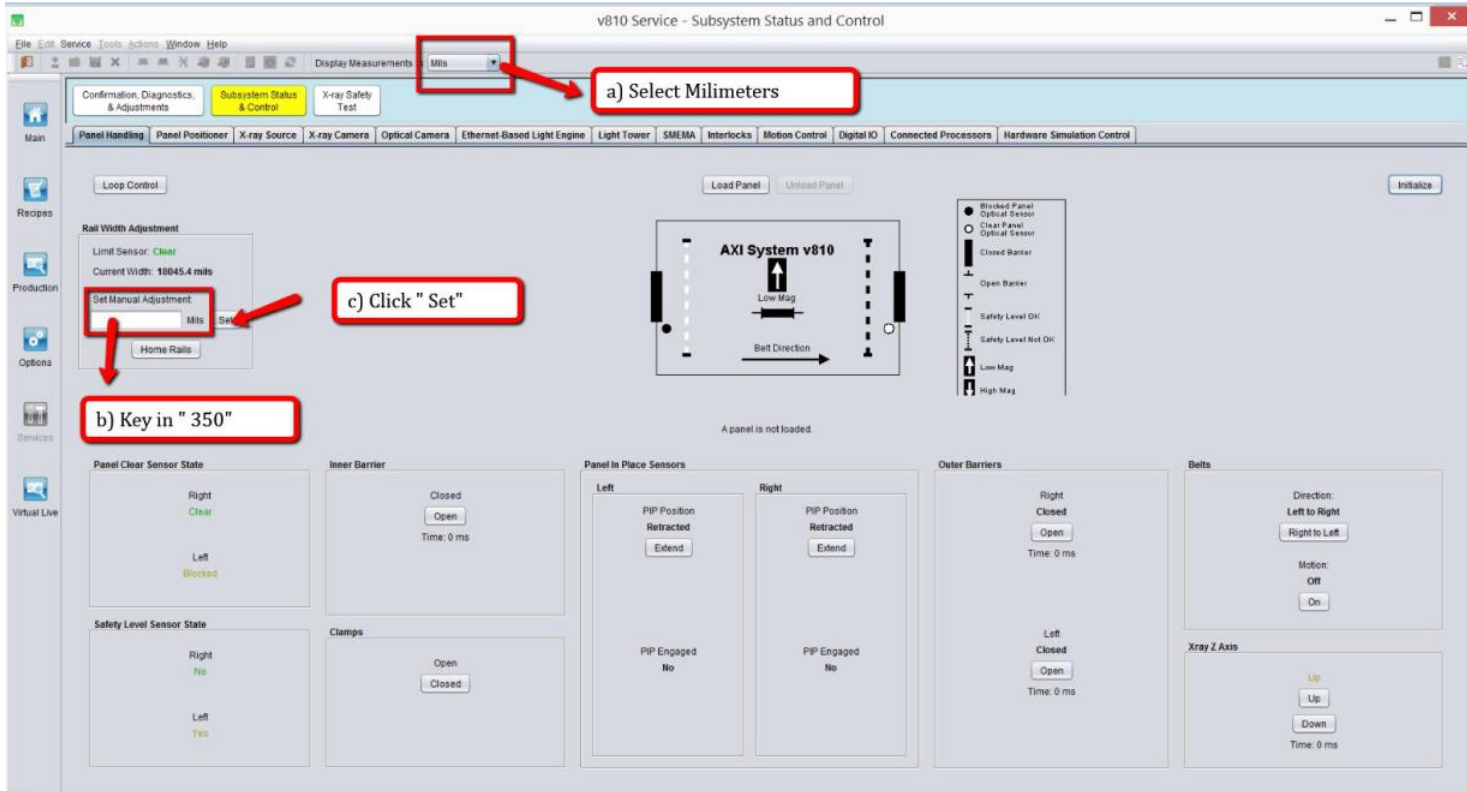


Figure 1 Panel Handling Interface

4. Wait until AWA is extend to desire width.
5. Close X-ray at Operator Console Panel and open the Front Door.
6. In advance, measure entire length of caliper.

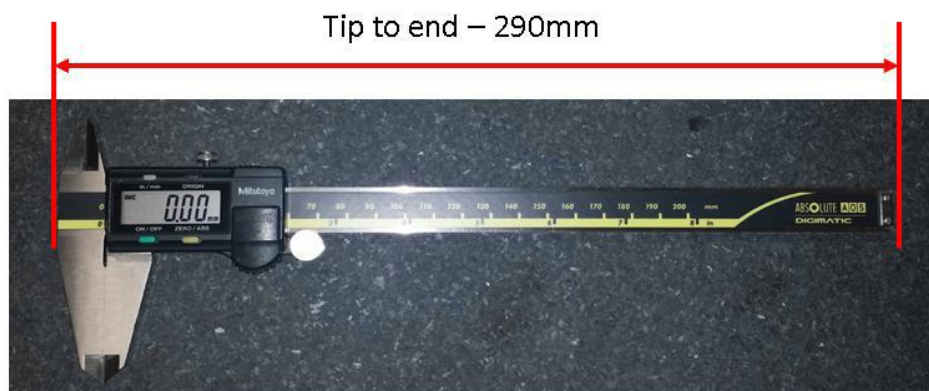


Figure 2 Caliper Length (Reference Pic Only)

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7. Place Caliper as below position. (May refer to below Figure 3)

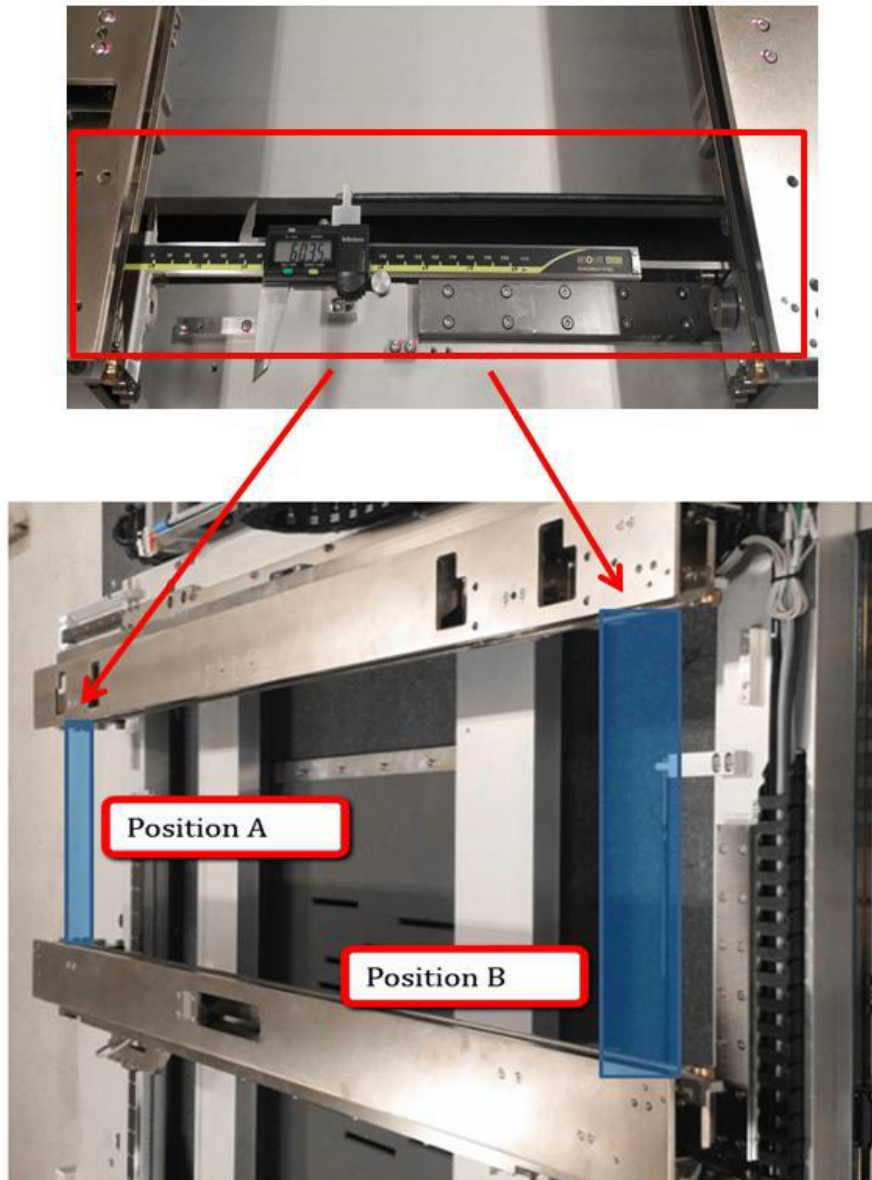
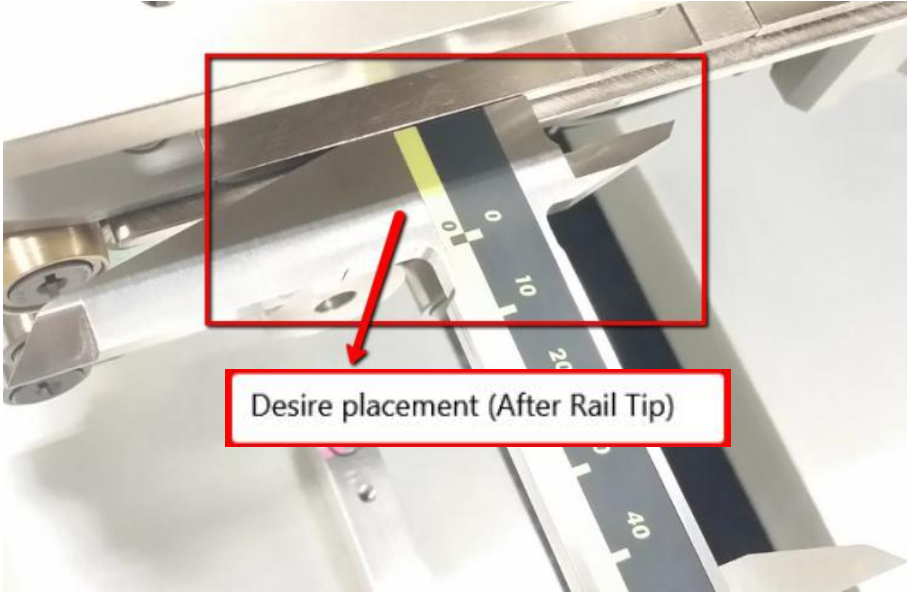
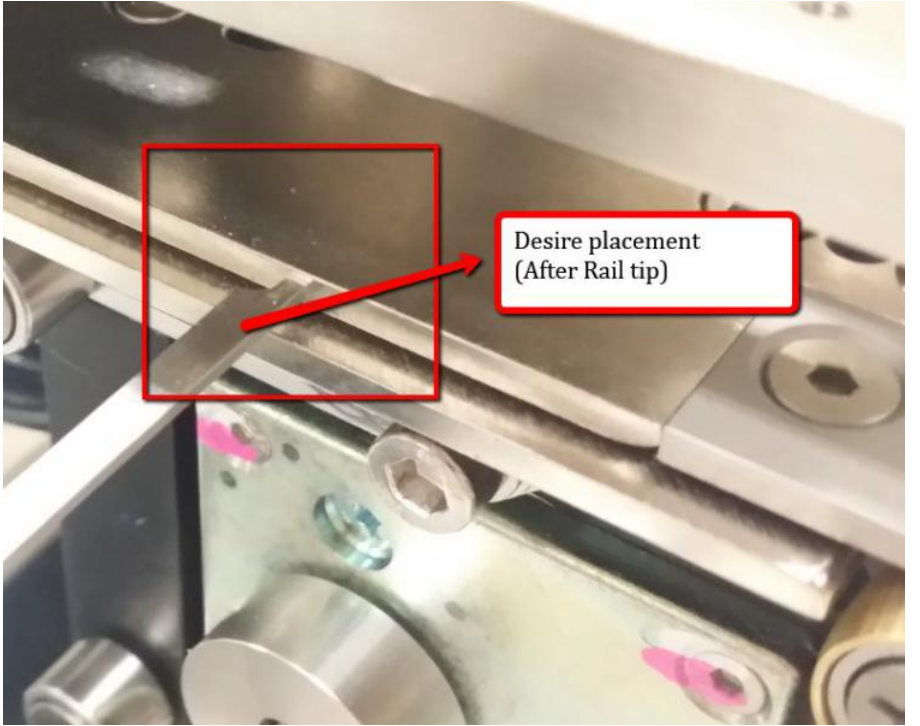


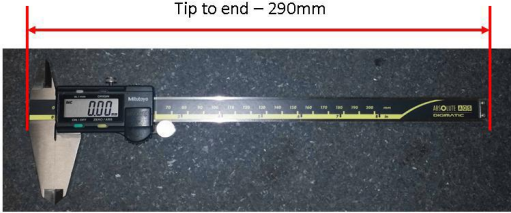
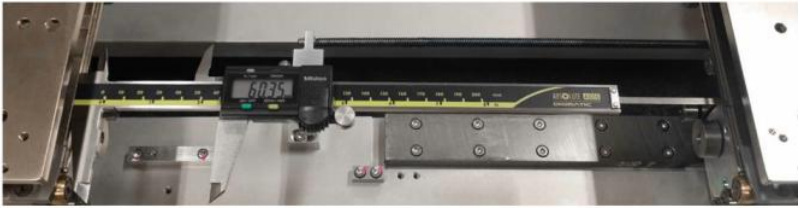
Figure 3 Caliper Placement

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Position	Position A & Position B
Fix rail Caliper placement	
Adj rail Caliper placement	

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8. Add up entire Caliper Length (Figure 2) with exact width of AWA

Item	Picture	Measurement
Exact length of Caliper		290
Exact Width of AWA		60.35mm
Total		350.35

9. Record 5 repeated measurements at the right end (Position A), and then 5 repeated measurements at the left end (Position B):

- MRave = average value of 5 repeated measurements at right end of conveyor
- MLave = average value of 5 repeated measurements at left end of conveyor

10. Calculate the parallelism, Mave and width error using the formulas below. Compare these values to the parallelism and width error limits.

- Parallelism = MRave - MLave
- Mave = (MRave + MLave)/2
- Width Error = Set Width - Mave
- Limits on Parallelism = +/- 0.125mm
- Limits on Width Error = +/- 0.125mm

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11. By refer to below table and step, check Parallelism Limit & Width Accuracy Limit

- If Parallelism Limit & Width Accuracy Limit is not within spec, please refer to [AWA Parallelism adjustment](#)

Sample Measurement (mm)			
Measurement Right End (Position A)		Measurement Left End (Position B)	
MR1	350.00	ML1	350.10
MR2	350.01	ML2	350.08
MR3	350.05	ML3	350.08
MR4	350.00	ML4	350.06
MR5	350.01	ML5	350.09

Calculations:

Measurement Right End average (MRave) = (MR1+MR2+MR3+MR4+MR5)/5 = 350.01

Measurement Left End average (MLave)= (ML1+ML2+ML3+ML4+ML5)/5 = 350.08

Measurement average = (MRave + MLave)/2 = 350.04

Compare to Limits for Parallelism:

MRave – MLave = **0.07** is less than Spec **" $\pm 0.125\text{mm}$ "**, so no adjustment for parallelism is required

Compare to Limits for Width Accuracy:

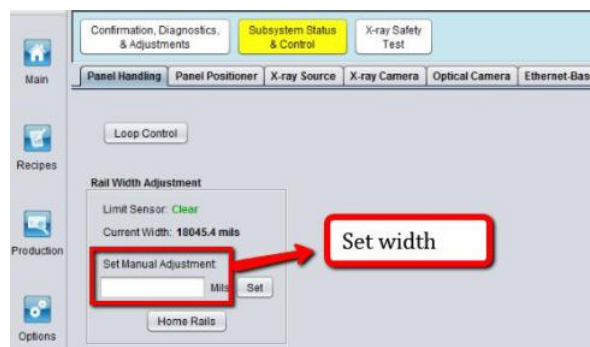


Figure 4 Set Width

Set Width – Mave = 0.04 is less Spec **" $\pm 0.125\text{mm}$ "**, so no adjustment required for width accuracy

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AWA Parallelism Adjustment

1. Click “Home Rails” at panel handling. AWA will extend to maximum opening.

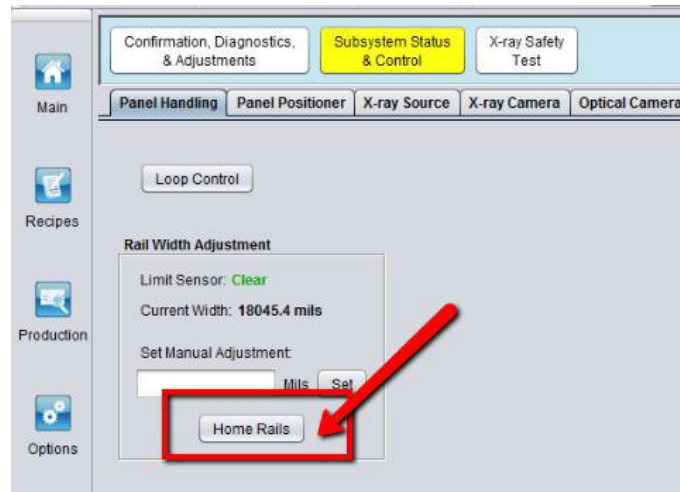


Figure 5 Home Rails

2. Open up front door, side door & rear door.
3. Use Size 8 hex wrench to loosen both side of jam nut while holding the set screw with Metrics Allen Key.

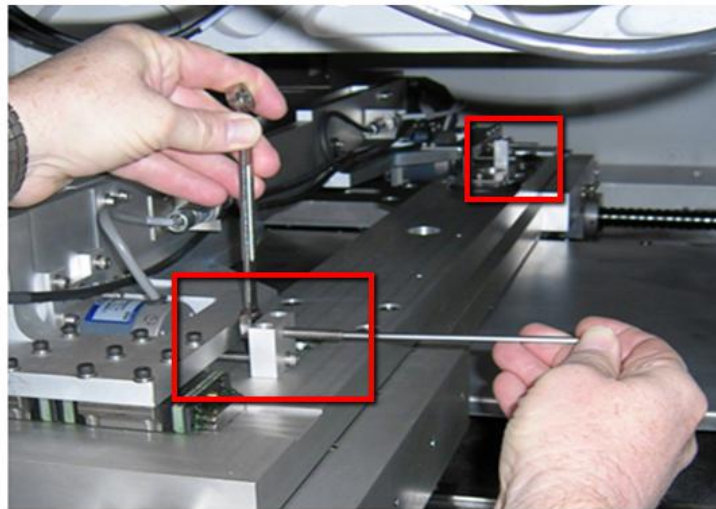


Figure 6 Jam Nut Loosen

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4. Insert 50 mil shim (4XACM-090) plate/feeler gauge in between of inner edge of the Slide Plate X-Axis and Set Screw. Once set screw tip in contact with Shim Plate, tighten up the jam nut.

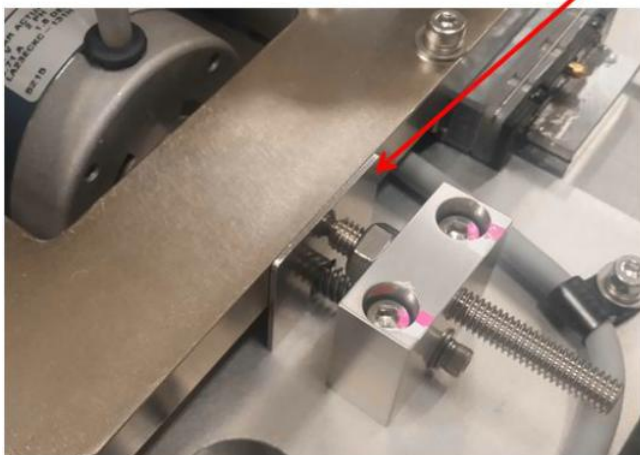


Figure 7 50 Mil Shim Plate Insert

5. Repeat steps 4 at other side of Adjustable rail.
6. Close SSCA.

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7. Place 2 pcs of XXL Auto Width Set Panel (4JACM-005) as below Figure 8. & 9. Manually push the Adjustable rail to allow both set panel fit in the desire width.

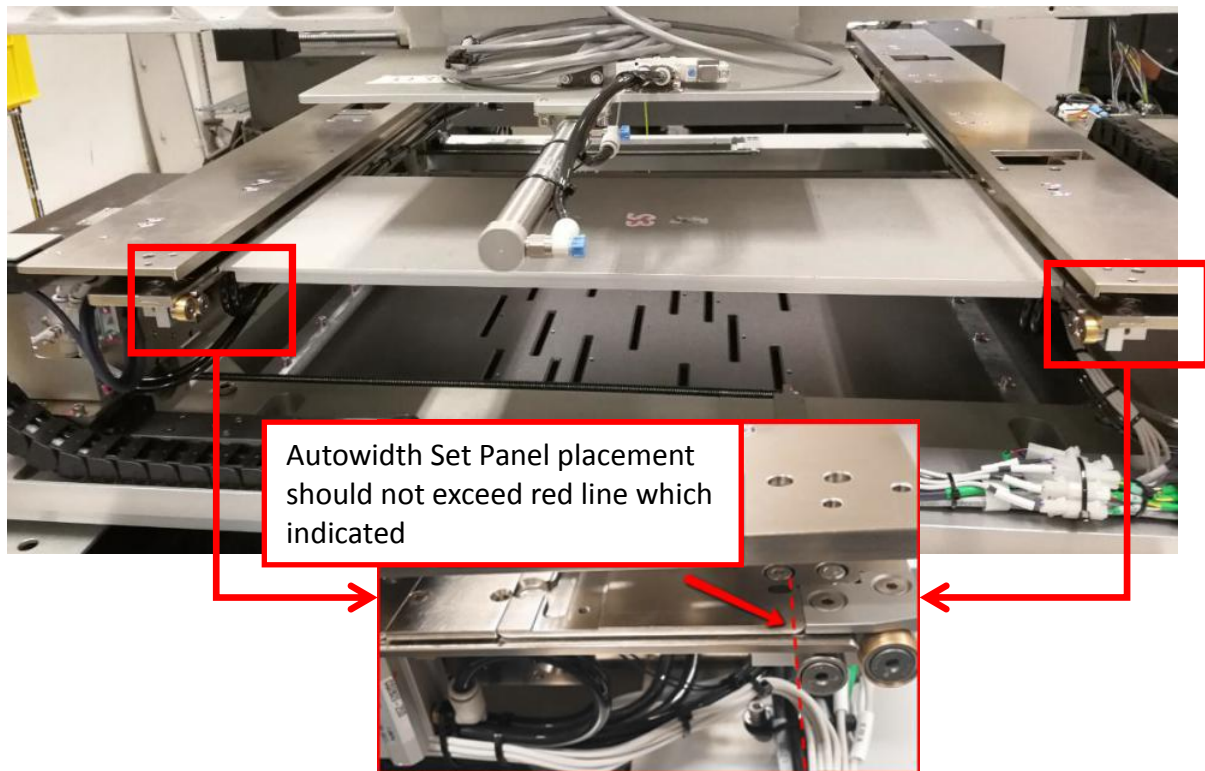


Figure 8 XXL Autowidth Set Panel (Left Side)

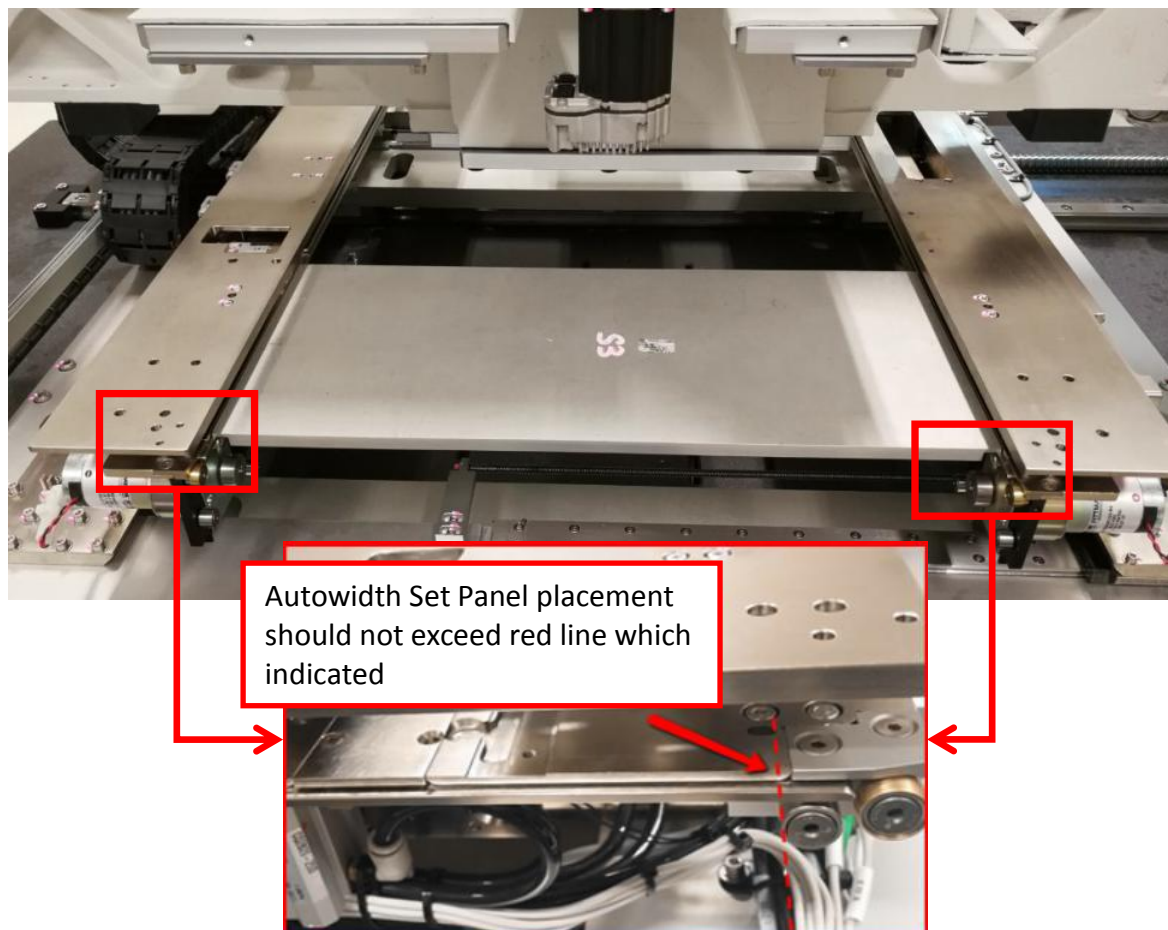
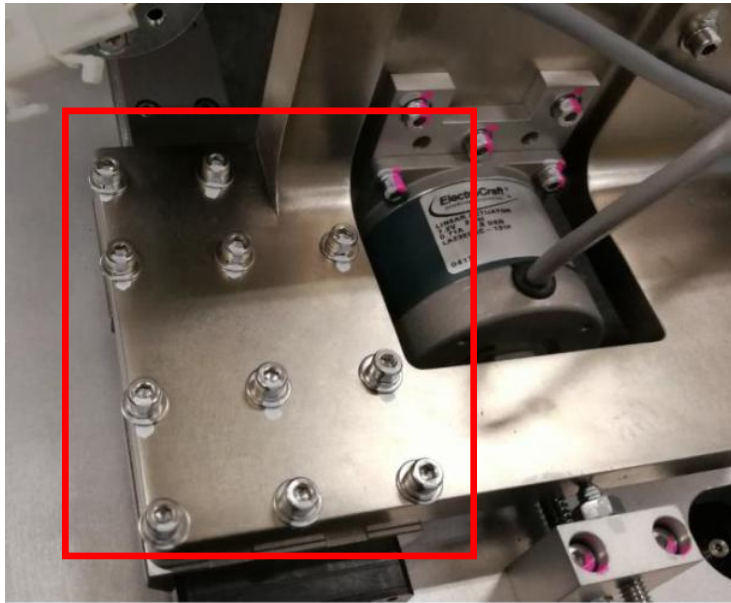


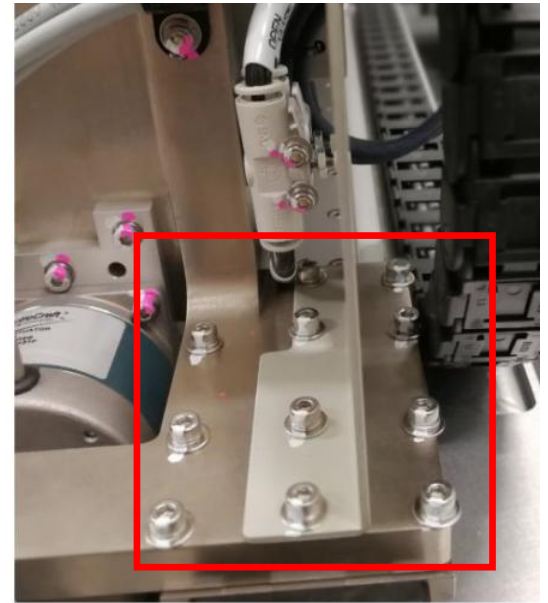
Figure 9 XXL Autowidth Set Panel (Right Side)

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8. Loosen screws at Adjustable Rail Linear Guide as below Figure 10.



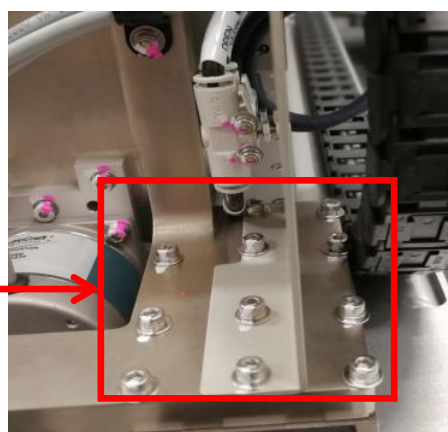
Right side of Adjustable Rail



Right side of Adjustable Rail

Figure 10 Linear Guide Screws

9. Push Left & Right Adj rail to allow Both Set Panel are fit into AWA. Tighten up both linear guide screws (Figure 11 & 12) at the same time.



Tighten up Left linear guide screws

Figure 11 Left Adj Rail Linear Guide alignment

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Tighten up Left linear guide screws

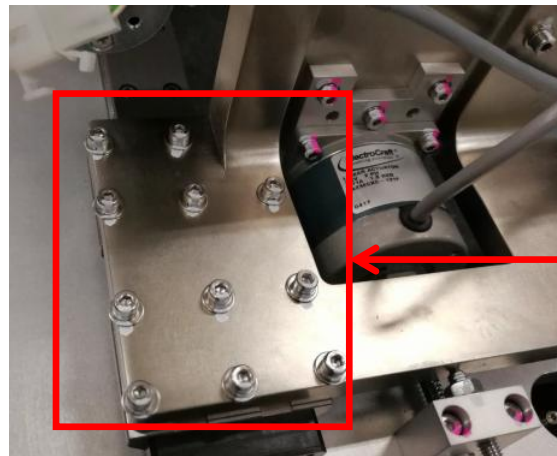


Figure 12 Right Adj Rail Linear Guide alignment

- Loosen 2 x Screws at AWA Home Sensor Bracket and Set the Rail Width Home sensor to be triggered (Red Colour)

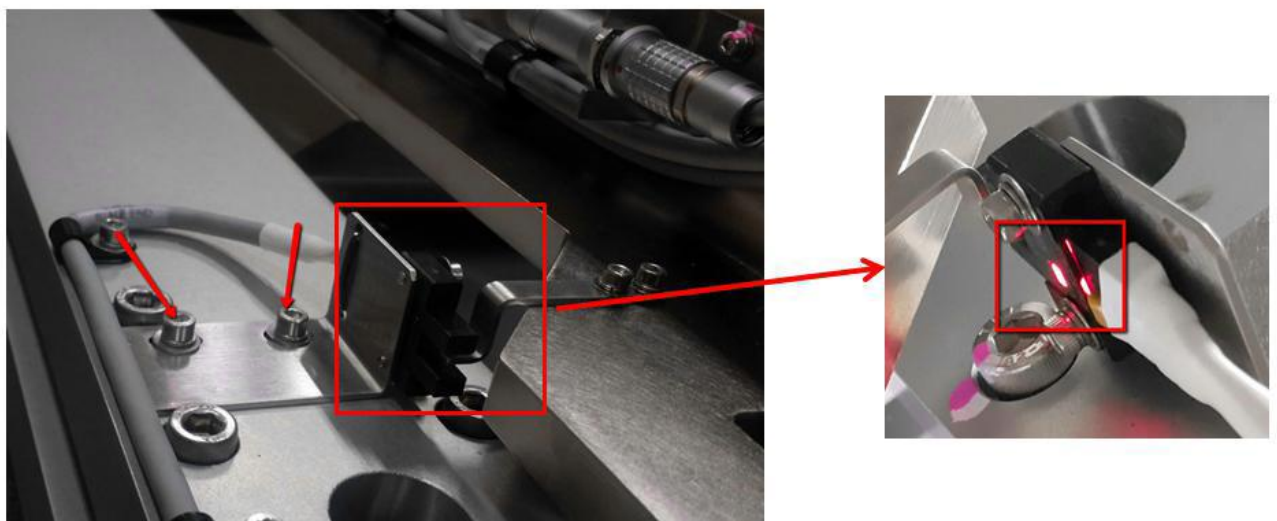


Figure 13 AWA Home Sensor Adjustment

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11. Once Rail Width Home Sensor triggered, gently pull away the bracket towards rear side approximately 1mm.

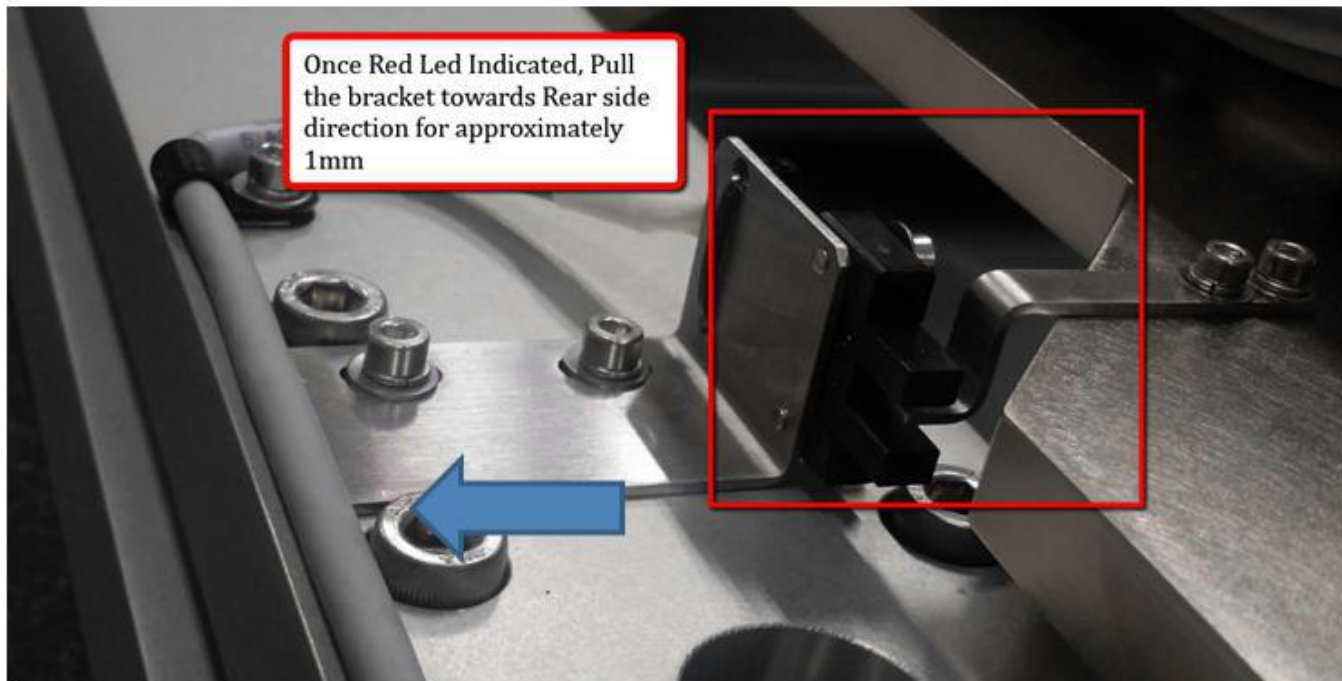


Figure 14 AWA Home Sensor Bracket adjustment

12. Tighten 2 x Screws at AWA Home Sensor Bracket.
13. Remove 2 x Autowidth Set Panel (4JACM-005) from rail.
14. Open back SSQA.
15. Go to Connector Processor Tab, Stop RMS & Restart RMS.
16. Initialize Digital IO.
17. Initialize at Panel Handling
18. Key in " 350 " mm at Panel handling and Click " Set "
19. OpenHardware Config by refer to below path "C:\Program Files\AXI System\AXI System\v810\5.0\s2exConfig
20. Search the key: railWidthAxisMaximumPositionLimitInNanometers =483186388(Default value) & record down. (Remarks : Every machine will consist varies of parameter, user will require to extract the parameter from particular machine)

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21. Use the caliper to measure AWA Rail Width & record down the actual measurement.

22. By refer to below table to calculate a new value of this parameter. (Below Value are for reference only)

Condition	Set value (Panel Handling)	Actual Measurement (Rail width)	Rail width parameter (HW config) – Step 20	Calculation Method
<i>If the measured value is greater than the expected value (rail width is too wide),</i>	350	355	483186388	$\text{New} = 483186388 + [(355 - 350) * 1\,000\,000]$ $\text{New} = 483186388 + (5 * 1\,000\,000)$ $\text{New} = 483186388 + 5\,000\,000$ $\text{New} = 488186388$
<i>If the measured value is less than the expected value (rail width is too narrow)</i>	350	345	483186388	$\text{New} = 483186388 + [(345 - 350) * 1\,000\,000]$ $\text{New} = 483186388 + (-5 * 1\,000\,000)$ $\text{New} = 483186388 - 5\,000\,000$ $\text{New} = 478186388$

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23. Open HW Config , enter new value at railWidthAxisMaximumPositionLimitInNanometers
24. Click Save at Note Pad.
25. Close V810 Software.
26. Open back V810 Software.
27. Initialize at Panel Handling.
28. Key in“ 350” mm at Panel handling and Click “ Set”
29. Open front access door, use caliper to measure AWA Rail Width & record down the actual measurement and compare with “Parallelism Limit & Width Accuracy Limit “
30. If still not within “Parallelism Limit & Width Accuracy Limit “, repeats Step 19~29.